GP Model V2: Statistical Audit

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GP MODEL TECHNICAL AUDIT (V2) | idata_gp_v2.pkl
Configuration: 3 Chains (Validated Subset) | Matérn 3/2 Kernel
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1. SAMPLING SPECIFICATIONS
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Warmup Iterations:  2000
Sample Iterations:  2000
Total Post-Warmup:  6000

2. MCMC DIAGNOSTICS
------------------------------
Max R-hat:          1.000
Total Divergences:  0
Numerical Status:   STABLE
Note: Chain 4 excluded due to numerical singularity.

3. PARAMETER ESTIMATES
------------------------------
                mean     sd  hdi_3%  hdi_97%  r_hat  ess_bulk
length_scale  0.259  0.036   0.194    0.324    1.0     927.0
variance      0.062  0.011   0.043    0.083    1.0    1526.0
b0           -7.551  0.044  -7.632   -7.469    1.0    1394.0

4. SPATIAL CHARACTERISTICS
------------------------------
Mean Length Scale:  0.2590
Effective Range:    0.4486 (Standardized units)
Spatial Variance:   0.0620
Moran's I (Resid):  -0.1693
======================================================================
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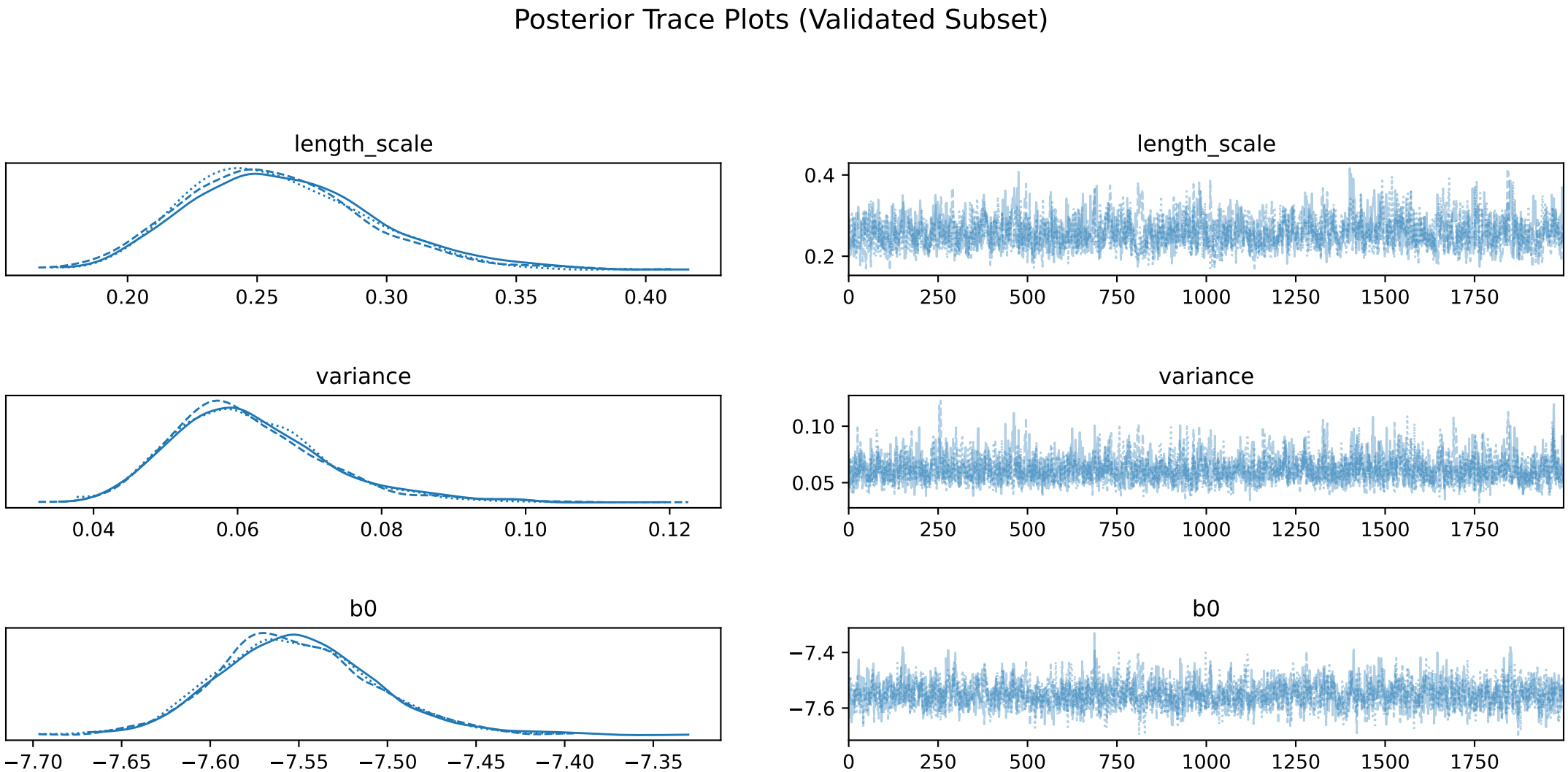
Posterior Trace Plots (Validated Subset)
length_scale
0.20
0.25
0.30
0.35
0.40
length_scale
0.4
0.2
0
250
500
750
1000
1250
1500
1750
variance
0.04
0.06
0.08
0.10
0.12
variance
0.10
0.05
0
250
500
750
1000
1250
1500
1750
b0
−7.70
−7.65
−7.60
−7.55
−7.50
−7.45
−7.40
−7.35
b0
−7.4
−7.6
0
250
500
750
1000
1250
1500
1750